

Sales Performance Analysis

Prepared for Alfido Tech | Intern Spark Data Analyst Project

Dataset: Kaggle — Superstore Sales (9,800 order lines, 4,922 orders, Jan 2015 – Dec 2018)

\$2.26M	4,922	\$459.48	793	+50%
Total Revenue	Total Orders	Avg Order Value	Unique Customers	Revenue Growth 2015 to 2018

Executive Summary

This report analyzes four years of Superstore transaction data (2015-2018) to evaluate sales performance across regions, product categories, and time. Revenue grew steadily from ~\$480K in 2015 to ~\$722K in 2018, with a strong and repeatable seasonal pattern peaking every November and bottoming every February. Technology leads on revenue on the strength of a small number of high-ticket items, the West and East regions outperform, and the South region lags. The dataset does not include Quantity, Discount, or Profit fields, so this analysis is scoped to revenue-based performance rather than profitability.

Methodology

Data was cleaned (11 missing **Postal_Code** values for Burlington, VT filled with the correct ZIP; dates parsed as day-first and verified by confirming every Ship_Date occurs on/after its Order_Date) and rolled up by order, category, sub-category, region, segment, and calendar month. KPIs, seasonality, and top/bottom performers were computed directly from the cleaned 9,800-row transaction table.

Trends & Seasonality

Revenue grew every year of the dataset: from ~\$480K (2015) to ~\$459K (2016, essentially flat), then ~\$600K (2017) and ~\$722K (2018) — roughly 50% cumulative growth. A clear seasonal cycle repeats each year: revenue bottoms out in February, climbs through the year, and peaks in November, with September and December also well above average.

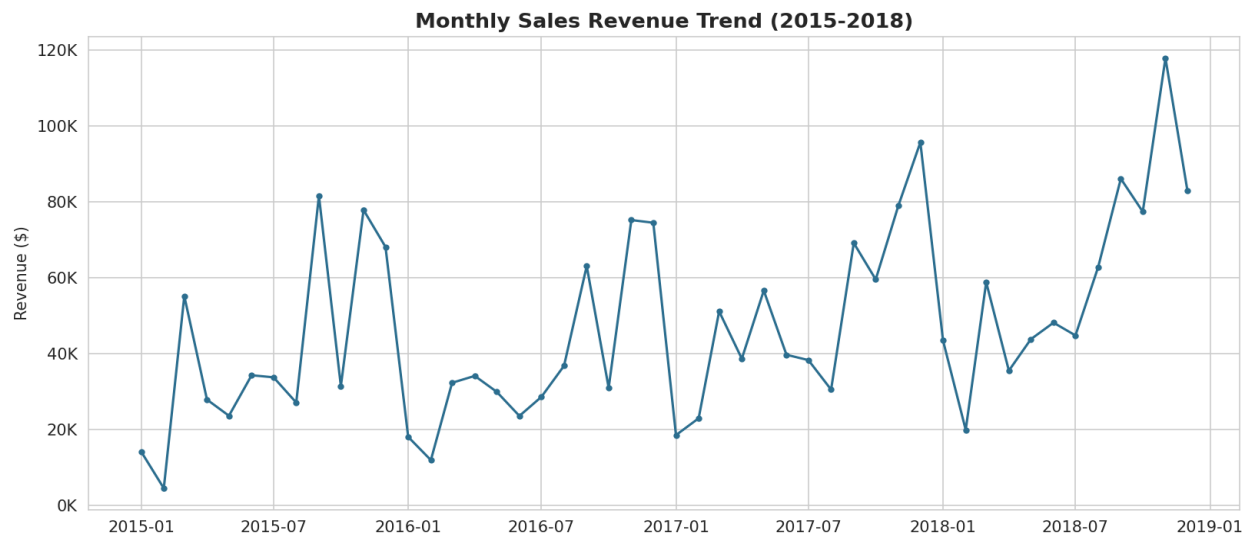


Figure 1. Monthly sales revenue, Jan 2015 - Dec 2018.

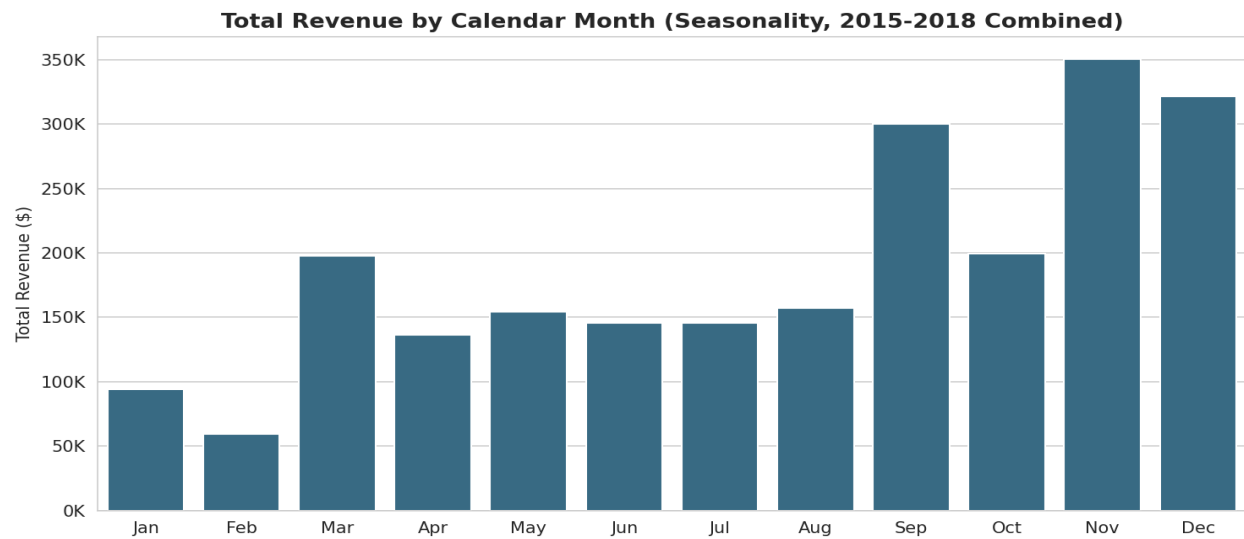


Figure 2. Total revenue by calendar month, all four years combined — the seasonal pattern.

Category & Product Performance

Category	Revenue	Orders	Line Items
Technology	\$827,456	1,519	1,813
Furniture	\$728,659	1,727	2,078
Office Supplies	\$705,422	3,676	5,909

Technology leads on revenue despite having the fewest orders and line items — its average sale size is much larger, driven by high-ticket copiers, video-conferencing systems, and large-format printers.

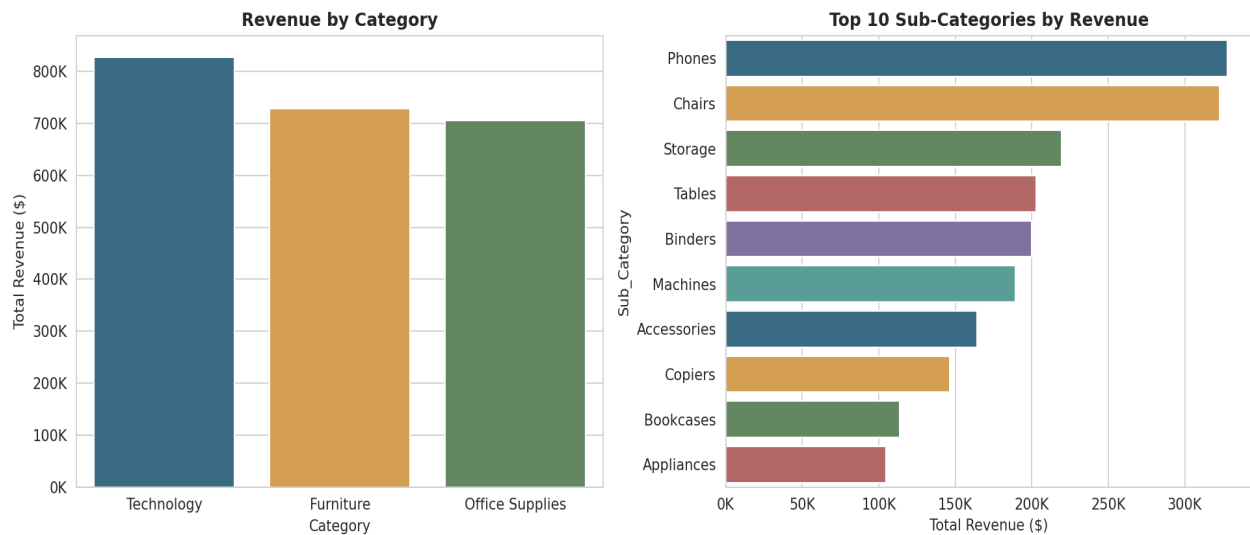


Figure 3. Revenue by category and top 10 sub-categories.

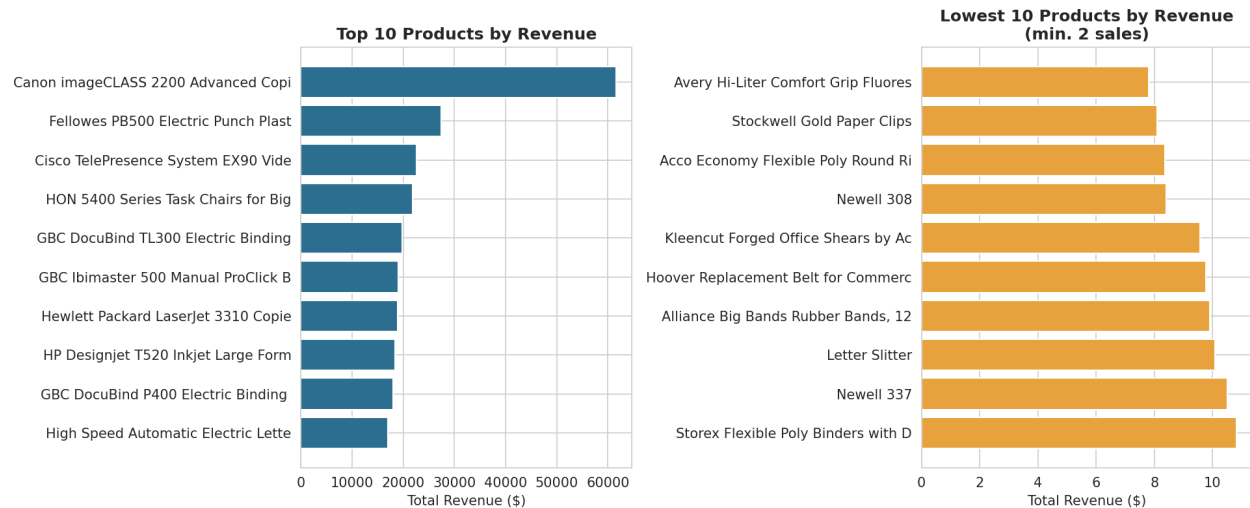


Figure 4. Top 10 and lowest 10 products by revenue.

Regional & Segment Performance

West and East regions generate the most revenue; South trails all other regions. California and New York alone generate more revenue than the entire South region combined. Consumer is the dominant customer segment at roughly half of total revenue, and Standard Class is the dominant shipping mode.

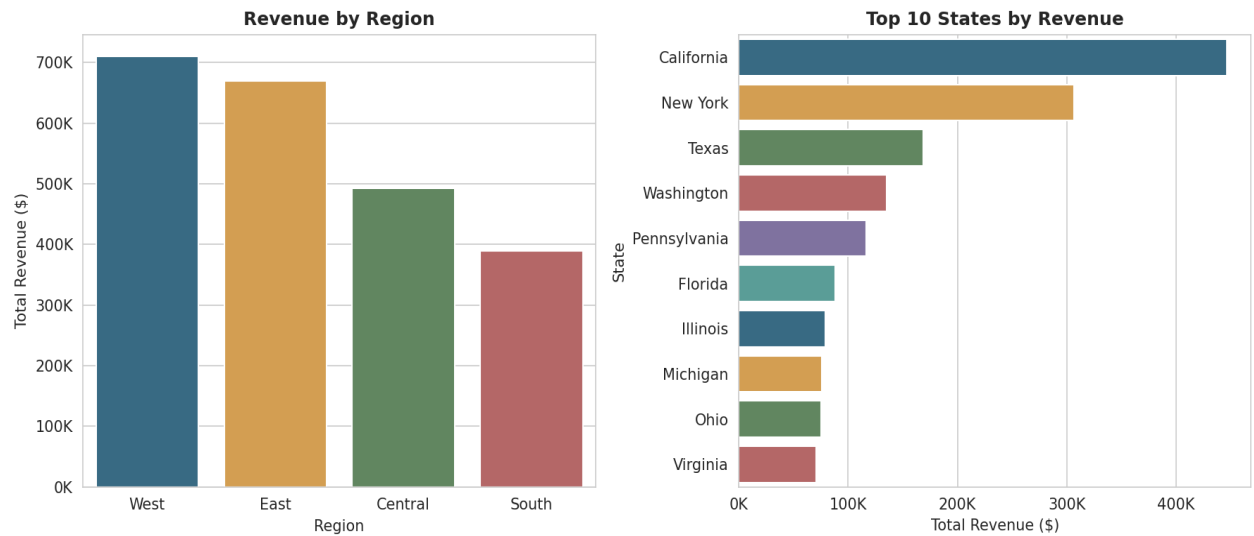


Figure 5. Revenue by region and top 10 states.

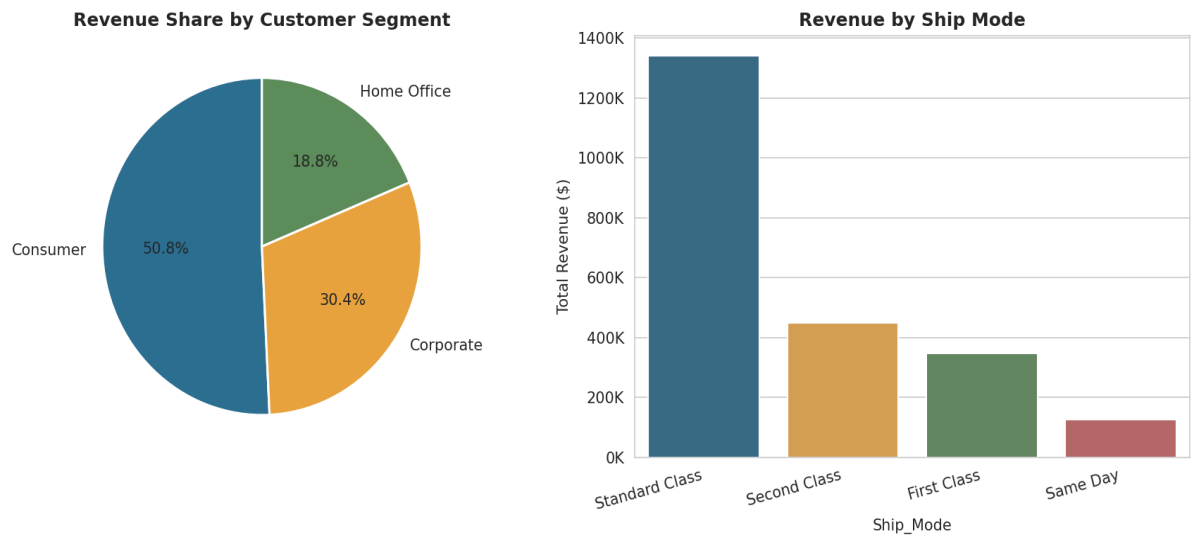


Figure 6. Revenue share by customer segment and by ship mode.

Key Findings

- Total revenue is ~\$2.26M across 4,922 orders (9,800 line items) from 793 unique customers, Jan 2015 - Dec 2018, with an average order value of ~\$459.
- Revenue grew every year, from ~\$480K (2015) to ~\$722K (2018) — roughly 50% growth over four years, though 2016 was essentially flat vs. 2015.
- Strong, repeatable seasonality: revenue bottoms out in February and peaks in November every year, with September and December also well above average.
- Technology leads on revenue despite fewer orders, driven by high-ticket items dominating the top-10 product list.
- West and East regions outperform; South lags, and California + New York alone outsell the entire South region.
- Consumer is the dominant segment (~51% of revenue), and Standard Class is the dominant shipping mode.
- The dataset has no Quantity, Discount, or Profit fields, so profitability and unit-economics questions cannot be answered from this data alone.
- Bottom-performing products are all low-price office accessories, forming a long tail of low-value SKUs.

Tactical Recommendations for Alfido Tech

1. Plan inventory and staffing around the Sep-Dec peak

November revenue runs roughly 6x February's. Pre-building stock for high-velocity sub-categories (Phones, Chairs, Storage) ahead of Q3/Q4, and running February-March promotions to smooth the post-holiday trough, would reduce both stockouts and idle-period revenue loss.

2. Double down on Technology's high-ticket items

Copiers, video-conferencing systems, and large-format printers punch far above their order-count weight. Bundling accessories/warranties with these items, or a targeted "business upgrade" push to Corporate and Home Office segments, can extend this strength.

3. Investigate the South region's underperformance

South trails every other region despite the same product catalog. Reviewing rep coverage, marketing spend allocation, or local competition in Southern states could reveal a fixable gap rather than an inherent demand ceiling.

4. Rationalize the long tail of low-value SKUs

Products like highlighters, paper clips, and rubber bands generate single-digit-dollar revenue per sale. Consider bundling these into multi-packs or with higher-margin items, and reduce marketing emphasis on them in favor of higher-revenue sub-categories.

5. Add profit and unit-level tracking to future data collection

This dataset only captures revenue — without Quantity, Discount, or Profit, Alfido Tech cannot tell a profitable steady seller from a high-revenue but heavily-discounted one. Capturing these fields going forward would unlock true margin-based decisions, not just topline revenue analysis.

Notes on methodology: Dataset covers 9,800 order line items / 4,922 unique orders / 793 unique customers, Jan 2015 - Dec 2018. Dates were parsed as day-first (DD/MM/YYYY) and verified against the shipping constraint. This dataset variant excludes Quantity, Discount, and Profit — all findings and recommendations are based on Sales (revenue), not profitability.

Prepared by: Data Analyst Intern, Intern Spark | Source notebook: Sales_Performance_Analysis_Alfido_Tech.ipynb